

7 Leaf scald is a disease of sugarcane caused by bacteria.

These bacteria are found in the **xylem** and release chemicals, one of which is called albicidin.

Some of these chemicals stimulate the xylem vessels to produce a sticky gum.

Albicidin inhibits chloroplast development.

The photograph shows the appearance of part of a sugarcane leaf infected with these bacteria.



- **white** streaks on leaf

(Source: © Nigel Cattlin / Alamy Stock Photo)

- (a) Explain how infection with these bacteria results in the death of the leaf cells and eventually the whole plant.

(4)

[illegible]

(b) Albicidin is an inhibitor of the enzyme DNA gyrase.

This enzyme is present in prokaryotic cells.

It binds to DNA and causes the DNA to supercoil, a process needed for cells to function properly.

- (i) Describe **two** differences in the structure of DNA found in prokaryotic cells and the structure of DNA found in human cells.

(2)

- (ii) Albicidin is being developed as an antibiotic to human pathogens.

Suggest why albicidin can be used as an antibiotic.

Use the information in the question to support your answer.

(3)

(iii) Explain the importance of developing new antibiotics, such as albicidin.

(2)

(Total for Question 7 = 11 marks)
